

COLLEGE INTERNSHIP MANAGEMENT SYSTEM

Mini Project Report

S.Y. BCS

Technology: Python, Flask, SQLite, HTML, CSS, JavaScript

Academic Year: 2026

Student Name: _____

Roll No.: _____

College: _____

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Assignment 1. Preliminary Investigation and its activities

1.1 Problem identification and definition

Students often depend on notice boards, messaging groups, spreadsheets or separate websites to find internships. The proposed system provides a centralized web platform for registration, internship search, online application and application tracking.

1.2 Problem Description

Internship information and application status may be distributed across different sources. The system stores students, internships and applications in SQLite and provides student and administrator interfaces.

1.3 Fact Finding techniques

Observation, informal interviews/discussions, document study, feedback from students and prototype testing were used to identify system requirements.

1.4 Drawbacks of Existing system

Scattered information; manual tracking; difficulty checking status; duplicate applications; inconvenient record maintenance; lack of a centralized interface.

1.5 Scope of the Proposed System

Student registration/login, dashboard, internship browsing, search, location filter, online application, duplicate prevention, status tracking, admin login, internship CRUD and application approval/rejection. Company login and resume upload are future scope.

1.6 Feasibility Study

Technically feasible with Flask and SQLite; economically low-cost; operationally simple for students/admin; suitable for the project schedule; production deployment should use stronger security.

Assignment 2. Requirement Specification

2.1 Data Requirements of the System

Students: id, full_name, email, college_name, course, password. Internships: id, company_name, internship_title, location, duration, stipend, skills. Applications: id, student_id, internship_id, application_date, status.

2.2 Identify End Users of the System

Student and Administrator are the implemented end users. Company login is not currently implemented.

2.3 Input Data to the System

Registration details, login credentials, internship details, search/filter values, application request and admin approval/rejection.

2.4 Output Information from the System

Dashboard details, internship cards, filtered results, application list, application status and admin statistics.

2.5 Functional / Nonfunctional / Processing Requirements

Functional: authentication, internship management, application management and status tracking. Nonfunctional: usability, reliability, maintainability, basic access control and performance. Processing: SQL retrieval/insertion/update, duplicate checking, joins, counts and client-side filtering.

Assignment 3. Database Design

3.1 Identify the entities and the attributes

Entities are STUDENTS, INTERNSHIPS and APPLICATIONS.

3.2 E-R Diagram

STUDENTS 1—M APPLICATIONS M—1 INTERNSHIPS. Applications connect students with internships.

3.3 Identifying all tables, fields, relationship between tables etc.

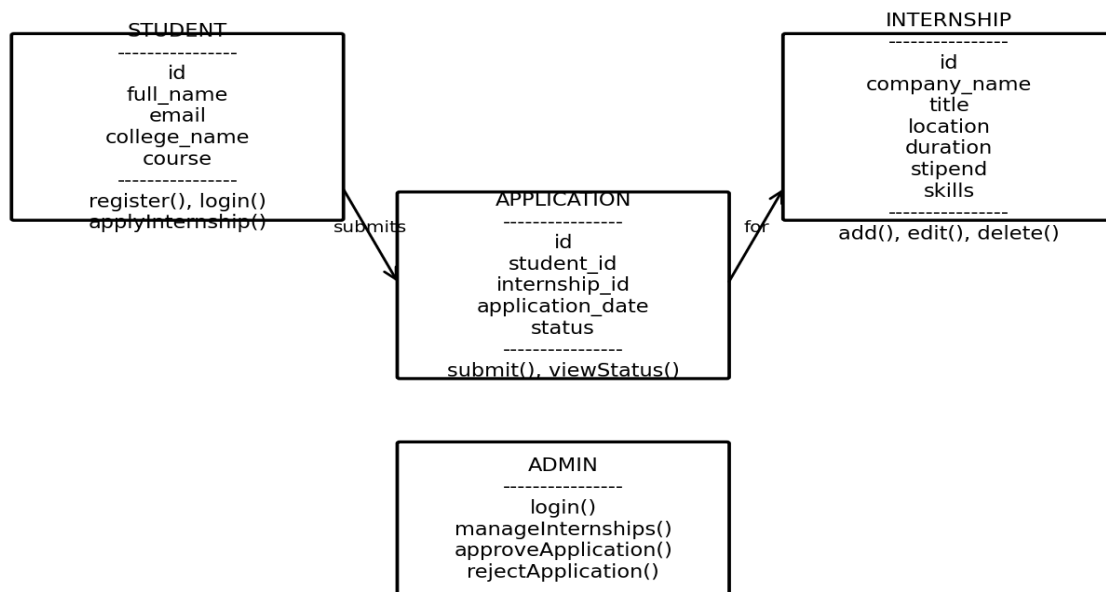
students(id PK, full_name, email UNIQUE, college_name, course, password); internships(id PK, company_name, internship_title, location, duration, stipend, skills); applications(id PK, student_id FK, internship_id FK, application_date, status).

3.4 Normalize database

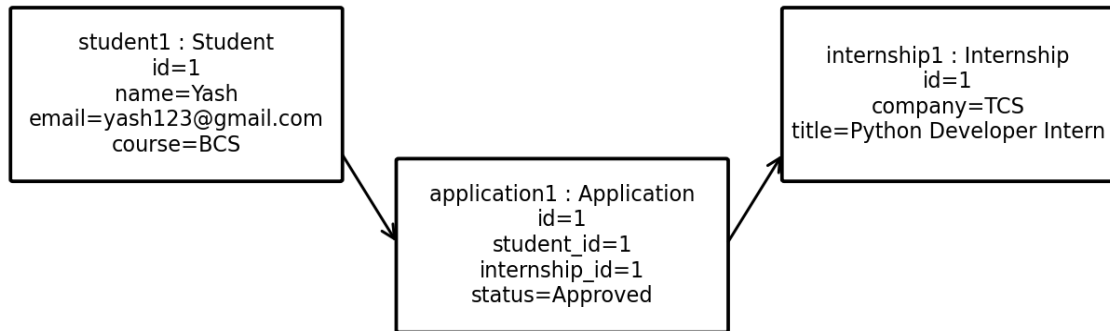
The design follows 1NF, 2NF and 3NF principles by separating student, internship and transaction data and avoiding repeated entity information.

Assignment 4. System Design

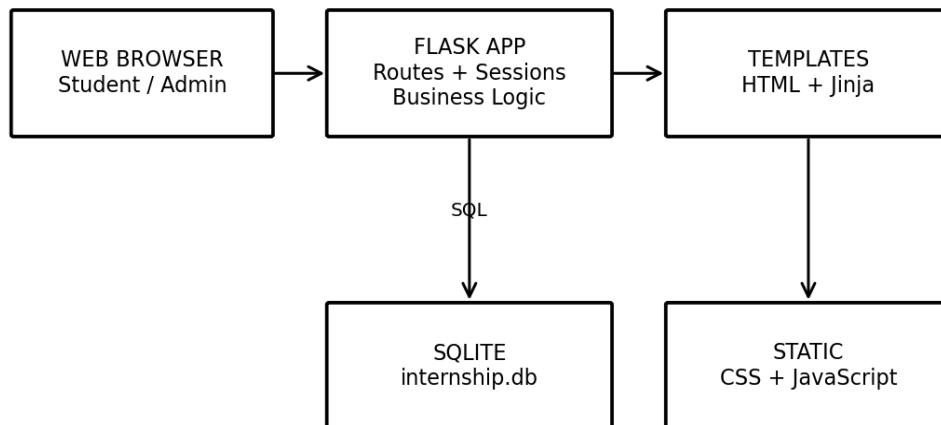
4.1 Class diagram



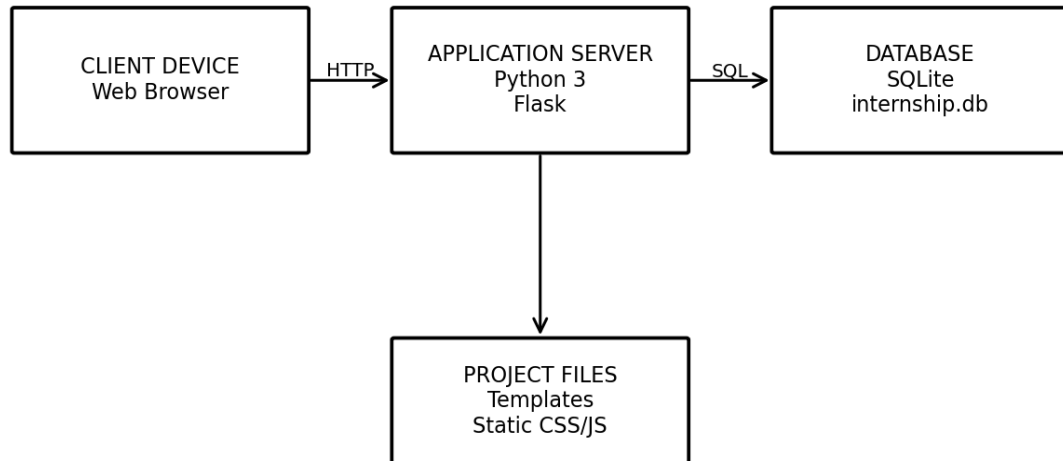
4.2 Object diagram



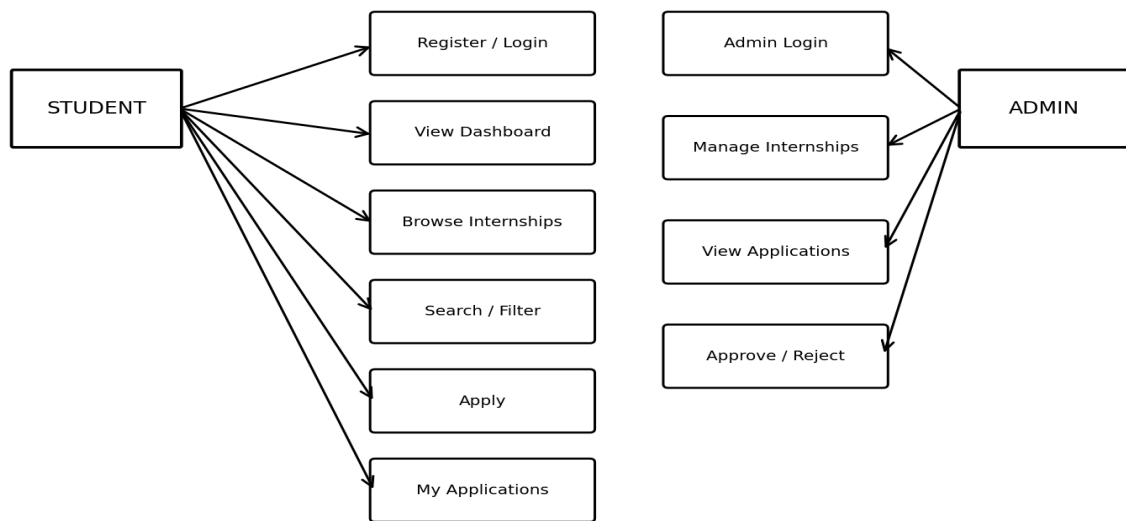
4.3 Component diagram



4.4 Deployment diagram

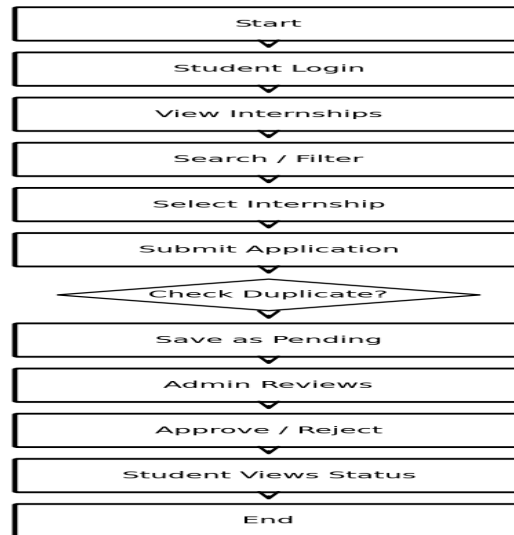


4.5 Use case diagram

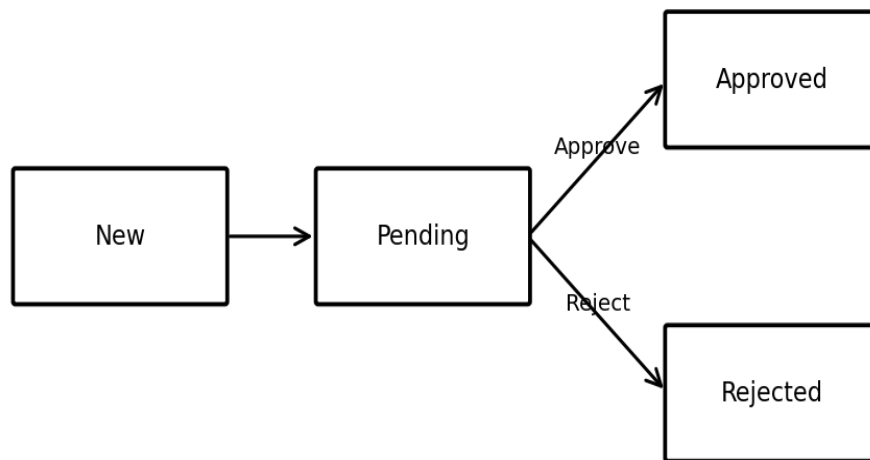


Student use cases include registration, login, dashboard, internship browsing, search/filter, application and status tracking. Admin use cases include login, internship management, application review and approval/rejection.

4.6 Activity diagram

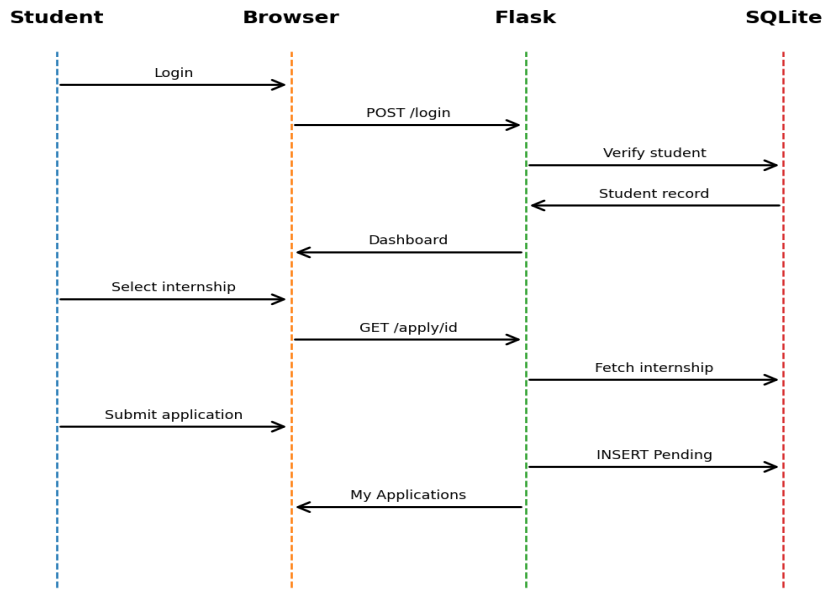


4.7 State chart diagram

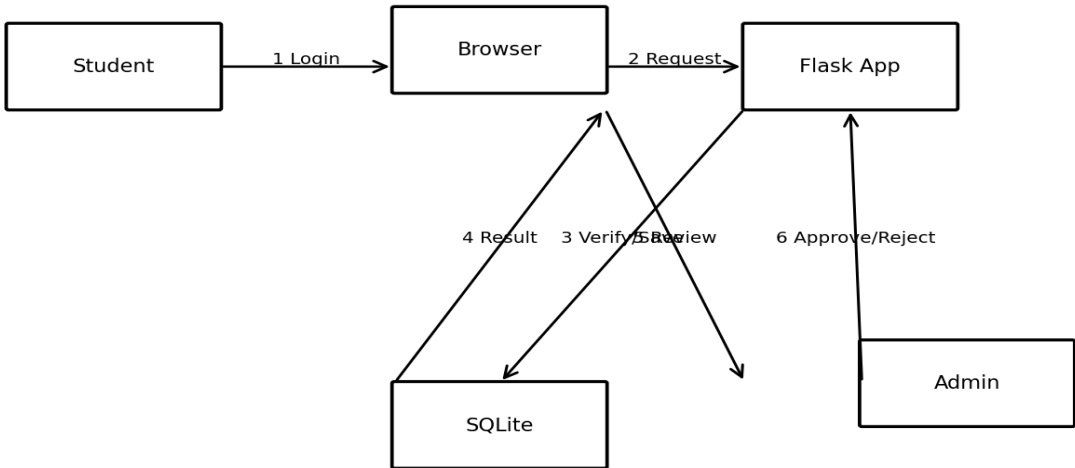


An application starts as New/Pending and reaches a final Approved or Rejected state after administrator action.

4.8 Sequence diagram



4.9 Collaboration diagram



FUTURE SCOPE

- Company registration and Company Login.
- Company Dashboard for managing internship opportunities and applications.
- Resume PDF upload by students with secure viewing/download for authorized users.
- Email or notification system for application status updates.
- Secure password hashing and stronger authentication.
- Database-based admin and role management.
- Internship recommendation based on student skills and interests.
- Cloud deployment and scalable database support.
- Advanced reports and analytics.

CONCLUSION

The College Internship Management System provides a centralized platform for internship discovery and application management. The implemented system covers student registration/login, internship browsing, search/filter, online application, application tracking, and administrator internship and application management. Company Login and Resume Upload are reserved for future enhancement.

REFERENCES

Python Documentation; Flask Documentation; SQLite Documentation; HTML, CSS and JavaScript documentation; Software Engineering and UML learning resources.